CPT205 Assessment 2

3D Modelling Project — A Futuristic Cityscape

Module	CPT205 Computer Graphics
Degree Programme	Information and Computing Science
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1. Brief Description

The title of my 3D Modelling project is **"Hi, Future!"**. Users can switch between 4 scenes using the keyboard to observe various areas of the future city, delve deep into the city, and interact with the elements and objects within it.

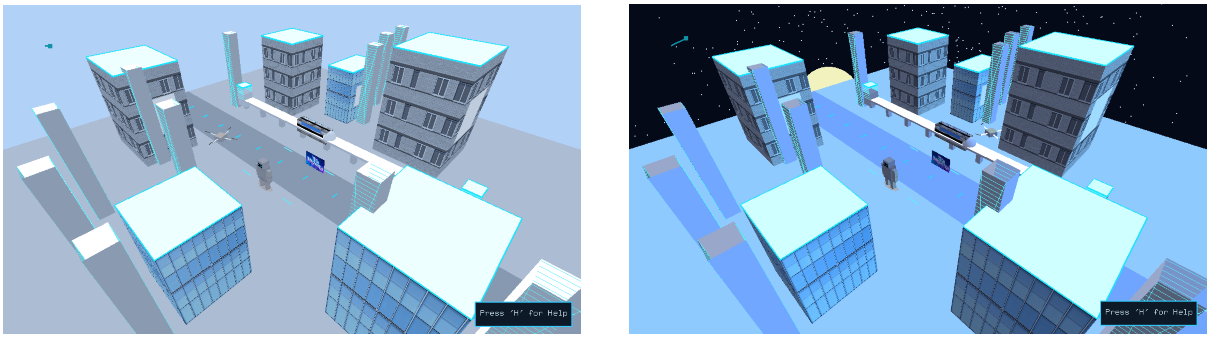

Figure 1: Cityscape Preview

The city landscape I created incorporates a variety of elements, including buildings, drones, trains, humanoid robots, etc. This system also supports dynamic day-night transitions, and users can adjust the lighting settings interactively to observe the real-time changes in lighting.

2. Features

2.1 Texture Mapping

Due to project limitations, I was unable to use additional graphics libraries. Therefore, in this project, I implemented a custom BMP texture loading process. By using the `ReadBMP()` function to read the BGR data of the .bmp file pixel. During the drawing stage, I used `glTexCoord2f()` to bind the texture coordinates with the model vertices, enabling the texture to be displayed in the scene at the correct position and direction. Additionally, since some textures have lots of details, when performing perspective transformation like in scene 3, the objects will occur a slight flickering phenomenon. To solve these problems, I enabled **multi-sampling** in the rendering pipeline. Compared with the traditional Mip mapping, multi-sampling can provide a smoother and stable texture rendering effect in the widely city view of this project. Finally, in order to enhance the users' experience of the system, I reversed the normal vectors and attached the texture to the interior of the sphere, creating a soft blue color sky in the daytime scene. And when adjusting the lighting, the top part can show a more natural color transition, further enhancing the overall realism of the scene.

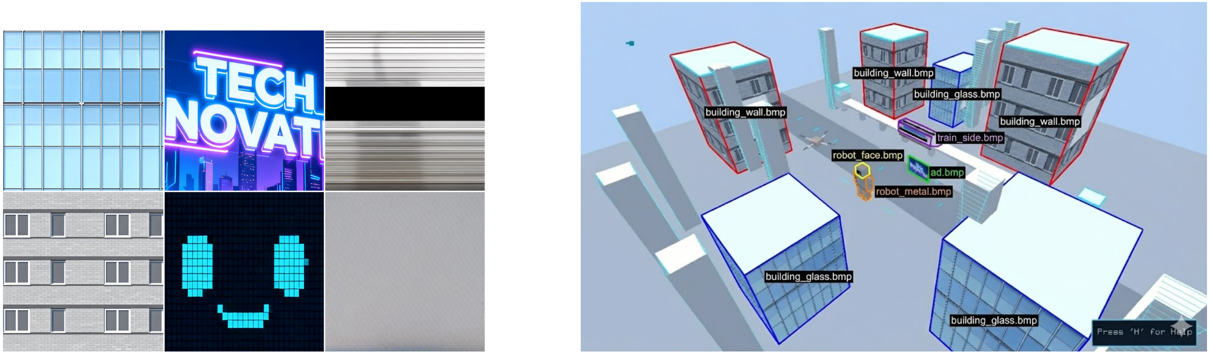

Figure 2: Texture and Texture used

2.2 Lighting and Material System

In this project, I used 3 different types of lighting. The **Directional Light** simulates sunlight and be used to achieve the overall illumination of the city. The robot eyes and train's light are applied by the **Point Lights**, enabling dynamic highlights on objects' surface. Moreover, the **Spotlights** is used for buildings' wall lamps, enhancing the sci-fi atmosphere of the environment.

In addition, I designed a diverse material properties to complement the lighting system, including metal surfaces, transparent and reflective glass, etc. These materials allow objects present coherent appearance and achieve the shadow effect. The lighting and materials used are shown in Fig. 3.

2.3 Hierarchical Modelling

I applied hierarchical modeling extensively in the construction of the Robot. The robot has complex structure of connected parts. By using `glPushMatrix()` and `glPopMatrix()` function, I established a parent-child relationship between the joints. For example, the torso moves the entire body, the upper arm rotation affects the lower arm, the thigh rotation affects the shin. All of these allow for complex posing of the robot. Similarly, I also designed Drone with hierarchical modeling where the propellers rotate independently relative to the drone body, which itself orbits the city center.

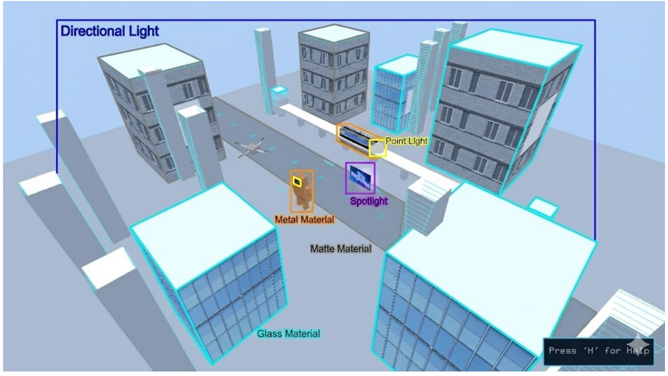

Figure 3: Lighting and Materials Applied

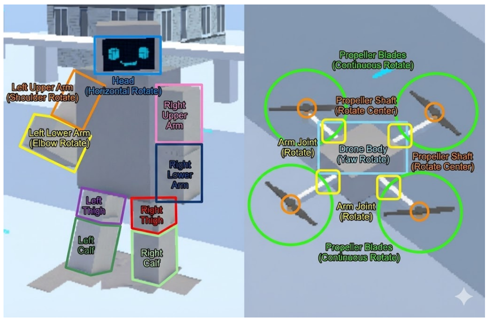

Figure 4: Hierarchical Applied

2.4 Animation System

The project achieves scenes continuous animation handled within the `idle()` function. The train moves along the tracks and enter and exit the tunnels. The drone can achieve the effect of circling the city or remain stop in the air. The joints of the robots can rotate. In addition, in the **scene 3**, I provided users with the function of moving the camera angle by rotating the view, so that they can have a comprehensive view of every corner of the future city.

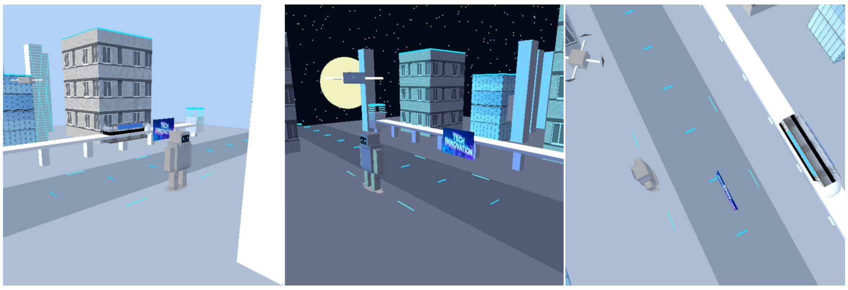

Figure 5: Scene3 Different Perspective Screenshots

2.5 Viewing and Camera Control

In this project, I designed 4 scenes to show different aspects of the city, the scene switches via number keys(1-4):

- **Scene 1 (Bird View):** From a high-angle lens position, users can see the layout of the entire city.
- **Scene 2 (Robot View):** The camera is focused and centered around the robot to facilitate users' observation of joint operations.
- **Scene 3 (Orbit View):** This scene applies an interactive camera that allows users to run around the street track by right click the mouse.

- **Scene 4 (First-Person Train):** The scene provides a dynamic camera. Users can press the "V" key to move the perspective to the interior of the train and follow the movement of the train model.

2.6 Additional Techniques

2.6.1 Planar Shadows

I used the shadow projection matrix to make the objects in the scene more realistic. By setting the scaling ratio of the y-axis to 0 to let the dark color cover the ground, which enhance the 3-dimensional effect of the objects.

2.6.2 Drone Interaction

The project used `gluProject()` function to achieve an interaction feature that when user click the drone in Scene 1, it can switch the camera to the drone's perspective.

2.6.3 Help Window

I have set a help window at the lower right corner of each screen to assist users in understanding the operation instructions for different scenes.

3. User Instruction

Keyboard Interaction[1]

- **Global Controls:**
  - `'1' - '4'`: Switch between Bird View, Robot View, Street View, and Train View.
  - `'B' / 'C'`: Switch between day and night mode.
  - `'H'`: Switch the Help Info window.
  - `Arrow Keys`: Adjust the direction of the lighting source.
  - `ESC`: Exit the program.
- **Robot Controls (Scene 2):**
  - `'Q'/'E'`: Rotate Head.
  - `'A'/'D'`, `'Z'/'X'`, `'J'/'L'`, `'N'/'M'`: Control Arm joints movements.
  - `'T'/'G'`, `'Y'/'R'`, `'U'/'K'`, `'I'/ O'`: Control Leg movements.
- **Train Controls (Scene 4):**

[1] Although for clarity, the report only lists the uppercase keys, the program accepts both uppercase and lowercase input with the same function.

- ’W’ / ’S’: Increase or decrease train speed.
- ’SPACE’: Pause/Resume train movement.
- ’V’: Switch first person perspective inside the train.

Mouse Interaction

- **Scene 1 (Bird View):** Left-click the flying drone to switch to the drone’s perspective.
- **Scene 3 (Street View):** Hold and drag the right mouse button to rotate the camera around the building.

4. Typical Screenshots

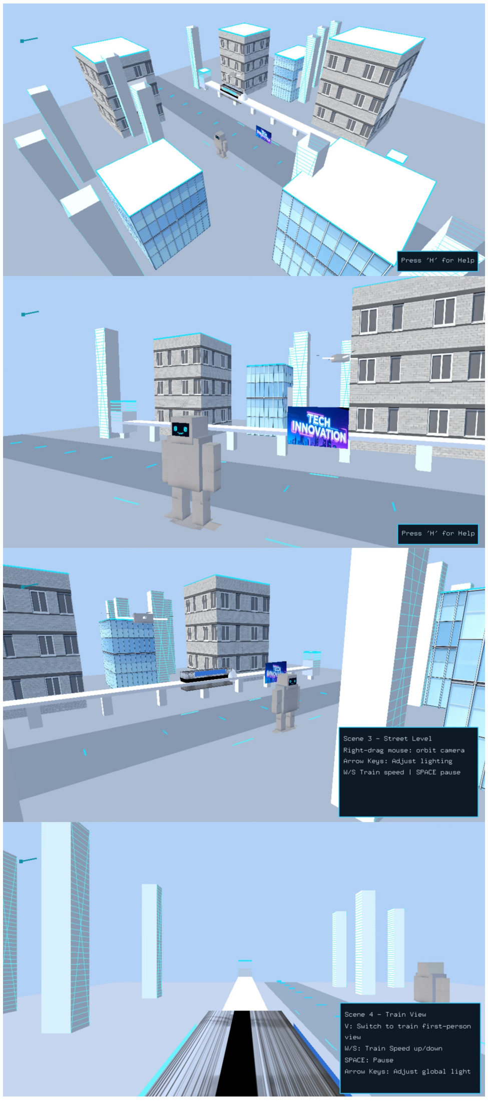

Figure 6: Day’s Scene Screenshots

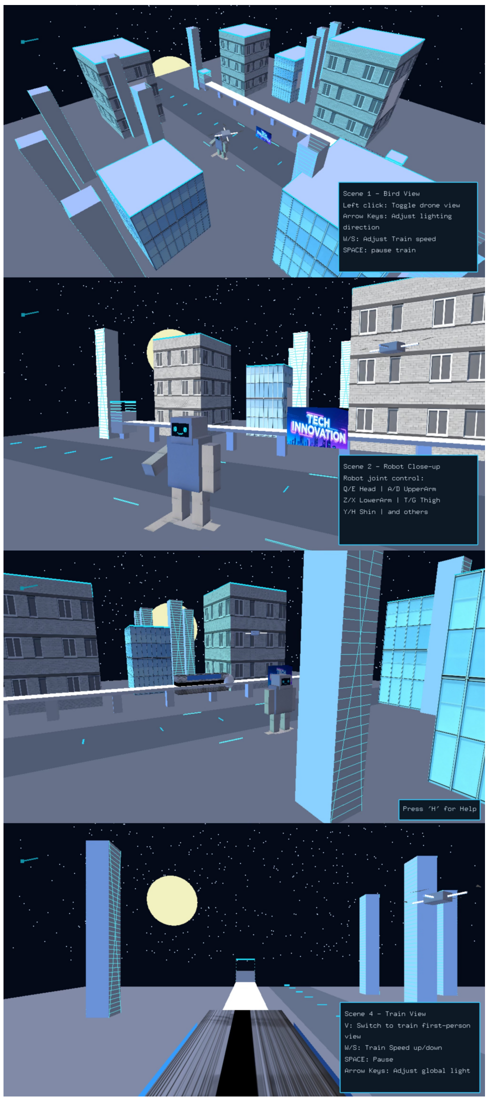

Figure 7: Night’s Scene Screenshots